Restoring Fish Passage in the Skeena Region - 2025

Executive Summary

SERNBC

SOCIETY FOR ECOSYSTEM RESTORATION IN NORTHERN BRITISH COLUMBIA

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Prepared for the Habitat Conservation Trust Foundation and the Ministry of Transportation and Infrastructure

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on behalf of the Society for Ecosystem Restoration in Northern British Columbia

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Executive Summary (PDF)

This report is available as a PDF and as an online interactive report at https://www.newgraphenvironment.com/fish_passage_skeena_2025_reporting/. We recommend viewing online as the web-hosted HTML version contains more features and is more easily navigable. Please reference the website for the latest version stamped PDF from fish_passage_skeena_2025_reporting.pdf.

Since 2020, the Society for Ecosystem Restoration Northern British Columbia (SERNbc) has been actively involved in planning, coordinating, and conducting fish passage restoration efforts within the Bulkley River and Morice River watershed groups, which are sub-basins of the Skeena River watershed. In 2022, the study area was expanded to include the Zymoetz River watershed group and the Kispiox River watershed group, followed by an extension in 2023 to encompass sections of the Kitsumkalum River watershed group, particularly where Highway 16 intersects the watershed.

The primary objective of this project is to identify and prioritize fish passage barriers within study areas, develop comprehensive restoration plans to address these barriers, and foster momentum for broader ecosystem restoration initiatives. While the primary focus is on fish passage, this work also serves as a lens through which to view the broader ecosystems, leveraging efforts to build capacity for ecosystem restoration and improving our understanding of watershed health. We recognize that the health of life - such as our own - and the health of our surroundings are interconnected, with our overall well-being dependent on the health of our environment.

Although the main purpose of this report is to document 2025 field work data and results, it also builds on reporting from past field activities and all reports can be considered living documents that can be updated and improved over time.

- Bulkley River and Morice River Watershed Groups Fish Passage Restoration Planning (2020)
- Bulkley River and Morice River Watershed Groups Fish Passage Restoration Planning 2021
- Bulkley River Watershed Fish Passage Restoration Planning 2022
- Skeena Watershed Fish Passage Restoration Planning 2022
- Skeena Watershed Fish Passage Restoration Planning 2023
- Skeena Watershed Fish Passage Restoration Planning 2024

Fish passage assessment procedures conducted through SERNbc in the Bulkley River, Kispiox River, Kalum River, Morice River, and Zymoetz River watershed groups since 2020 are amalgamated online within the Assessment Data Summary appendix of the report found here which includes links to project reporting for each site.

Environmental DNA (eDNA) sampling was the substantive field programme this season. A total of 19 samples were collected across 10 streams, alongside 1 field and office blanks run as protocol controls.

Of the 5 species assayed, rainbow trout (9 of 13 sites), bull trout (7 of 18 sites), and coho salmon (3 of 19 sites) were detected. chinook salmon and sockeye salmon were not detected in any sample. Detections use the four-droplet call threshold applied by the laboratory. A single season of sampling establishes presence, not absence — a species may go undetected because it is absent, or because it was not shedding detectable DNA at the place and moment sampled — which is why the value of this layer grows with repetition.

Monitoring was conducted at two sites, including post-remediation evaluations at Waterfall Creek (PSCIS crossing 124421) and Peacock Creek (PSCIS crossing 197962). Monitoring included eDNA sampling and habitat observations. A custom effectiveness monitoring form tailored to fish passage projects was used and metrics assessed included flow velocity, substrate condition, channel constriction, riparian condition, and cover availability.

A climate departure analysis was completed for the study area, comparing the recent decade against a pre-warming reference period of 1951 to 1980. Mean annual air temperature is 2.1 °C above that reference, and the warming is strikingly even across the region — the 6 ecoregions spanning the study area differ by only 0.28 °C in warming accumulated since 1951. The clearest signal is in the snowpack, and it is one of timing rather than of quantity: the snowmelt midpoint now falls 15 days earlier in the year and summer snow water equivalent has fallen 60 per cent. The freshet that crossing structures were sized against is arriving substantially earlier than the record they were designed from, and the late-summer cold water that snowpack once supplied is declining. Because the warming is so uniform, the thermal outlook for a given barrier is a question of elevation and local reach conditions rather than of which watershed group it sits in.

Functional floodplain extent was mapped for the Bulkley River watershed group, the first of the study area's watershed groups to be delineated. Modelled floodplain covers 48,344 ha, or 6.2 per cent of the 7,762 km² group. Passage and floodplain condition are the same problem viewed from two angles — reconnecting a stream longitudinally achieves less where the floodplain is severed laterally — so the layer indicates where remediation should account for more than the structure itself. The delineation is published to a shared spatial catalogue and read into this report rather than regenerated for it, so one modelled product serves every report that needs it; Morice and Kispiox are already published and can follow without re-running the model.

The components of this initiative are designed to complement one another, building the state of knowledge regarding watershed health and directing restoration effort to where it will do the most good. Passage assessments and the provincial model establish where fish can reach; habitat confirmations, fish sampling and environmental DNA establish where they are; floodplain delineation and aerial imagery establish how infrastructure has altered the valley bottom; climate departure and water temperature establish the trajectory; effectiveness monitoring tests whether the work delivered. Together they make explicit what is known, what is not, and what should be measured next. Making that evidence usable is part of the work — data collected on structured digital forms, methods documented alongside the code that implements them, layers published to open catalogues, and reporting written so partners, First Nations, regulators and tenure holders can question it and build on it — so understanding accumulates across the program rather than being rebuilt each season.

A major challenge in advancing fish passage restoration is the complexity of working across jurisdictions and with multiple stakeholders—rail and highway authorities, forestry ministries, licensees, and private landowners. These partners are often being asked to accommodate priorities that originate outside their mandates and budgets. Convincing them to invest in difficult, high-cost interventions—like modifying crossings or relocating infrastructure—requires navigating uncertainty about costs and ecological outcomes, as well as a disconnect between the benefits to watershed health and the internal pressures or performance goals of these agencies. It's a tough ask: to take on massive, uncertain projects when they're already stretched thin with their own responsibilities.

Fish passage restoration across British Columbia is further complicated by the legacy of infrastructure deeply embedded in the landscape. Roads, railways, highways, community infrastructure and private assets often constrain floodplains and disrupt natural hydrological processes. While targeted repairs to individual barriers are essential, they won't resolve the broader systemic issues without rethinking and restructuring how infrastructure interacts with watershed function. Loss of riparian vegetation and intensive beaver management only add to the degradation. Addressing these challenges means making strategic, well-communicated choices, building trust, and staying committed to a longer-term transformation.

A single definitive ranking of remediation priorities is not achievable, because the criteria that determine whether a crossing is worth remediating are not measured in common units, and the deciding constraint sits outside the project in the capacity and willingness of infrastructure owners and tenure holders to support implementation across multi-year timelines. What can be done systematically is to maintain a well-characterised set of candidate sites, so that when a tenure holder is in a position to act, the ecological value of the opportunity and the appropriate scope of work are already understood.

Looking ahead, priorities for the program are:

- Continue and extend effectiveness monitoring across the region's remediated crossings, spanning a range of structure types, treatments and stream sizes, and continue developing a cost-effective framework that ties baseline and post-remediation data to measurable productivity gains.
- Continue environmental DNA sampling as an annual, region-wide layer so single-year detections become distribution and trend information, and resolve species use through targeted fish sampling where habitat ranks high but detections are absent or unresolved.

- Extend functional floodplain mapping from the Bulkley to the remaining watershed groups, and carry climate departure and water temperature into prioritization, so watershed condition and trajectory sit alongside barrier status wherever ranking is done.
- Resume Phase 1 assessments and continue calibrating the provincial habitat model against observed conditions, so the candidate set that prioritization draws on keeps growing and improving.
- Commission engineering designs at the Ministry of Transportation and Infrastructure crossings identified in previous reporting, and advance the highest ranked sites assessed this year.
- Continue developing the shared data and reporting infrastructure — open spatial catalogues, a public web map, and common repositories for water temperature and environmental DNA — so results are findable and reusable by everyone working in these watersheds.
- Continue working with First Nations, stewardship groups, educational institutions and tenure holders, feeding monitoring results back into prioritization and design as adaptive management.